

String Theory Seminar

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Abstract

This is the lecture note of the String Theory Seminar in EPFL 2026 spring semester. The main reference will be Polchinski's String Theory.

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1 Preliminary: Functional Formalism

In this section, we will discuss a functional formulation of quantum theory, which is the basis of the path integral approach to quantum field theory. The contents mainly follow the book [1] and [2].

In quantum field theory we mainly care about the transition amplitude between states and with operators inserted, which are called the correlation functions. Among all correlation functions, we mainly care about transition amplitudes between ground states with operators inserted:

$$\langle 0 | \mathcal{T} O(x_1) \dots O(x_n) | 0 \rangle \quad (1)$$

It is proven that the correlation functions are related to the scattering amplitudes of particles, through the LSZ reduction formula. Therefore, the correlation functions are one of the main objects of interest in quantum field theory.

Apart from the canonical quantization approach, there exist another approach in quantum theory, which makes the calculation of correlation functions more convenient. This approach is called the path integral approach, which is the main topic of this lecture.

1.1 Quantum Mechanics Path Integral

To formulate the path integral approach, we start with one dimensional quantum mechanic system with a standard non-relativistic Hamiltonian:

$$H = \frac{p^2}{2m} + V(x) \quad (2)$$

and focus on the standard translation amplitude between two position eigen states, then we generalize it to case with operator insertions and transition amplitude between vacuum states, and finally we will generalize it to quantum field theory.

1.1.1 Translation Amplitude as Functional Integral

We try to calculate the translation amplitude between two position eigen states $|x\rangle$ and $|x'\rangle$, by definition the translation amplitude is:

- **Schrodinger Picture** : with $|x\rangle$ as eigen state of position operator, we have:

$$\langle x' | e^{-iHt} | x \rangle \quad (3)$$

- **Heisenberg Picture** : with $|x, t\rangle$ as eigen state of position operator $x(t)$ at time t , we have:

$$\langle x', t | x, 0 \rangle \quad (4)$$

These two definitions are equivalent, its only a matter of choice of picture.

Remark

In fact we can prove that $|x, t\rangle = e^{iHt}|x\rangle$, thus directly see the equivalence of the two definitions.

The discussion in this subsection is straightforward in schrodinger picture, thus we will mainly work in schrodinger picture in this subsection.

We want to transform this operator expression into a functional integral, thus we need to insert complete set of states to make operators into c-numbers. The trick is as follows:

- **Step 1**: divide time interval into N small intervals $t = \sum_{i=1}^N \delta t$ and then insert complete set of states in each small interval:

$$\begin{aligned}
\langle x' | e^{-iHt} | x \rangle &= \langle x' | e^{-iH\delta t} e^{-iH\delta t} \dots e^{-iH\delta t} | x \rangle \\
&= \int \prod_{i=1}^{N-1} dx_i \langle x' | e^{-iH\delta t} | x_{N-1} \rangle \langle x_{N-1} | e^{-iH\delta t} | x_{N-2} \rangle \dots \langle x_1 | e^{-iH\delta t} | x \rangle
\end{aligned} \tag{5}$$

- **Step 2:** evaluate each infinitesimal translation amplitude $\langle x_{i+1} | e^{-iH\delta t} | x_i \rangle$ we notice that:

$$e^{-iH\delta t} = e^{-i\frac{p^2}{2m}\delta t} e^{-iV(x)\delta t} e^{O(\delta t^2)} \tag{6}$$

Remark

The above formula is a consequence of the Baker-Campbell-Hausdorff formula, which states that for two operators A and B , we have:

$$e^{A+B} = e^A e^B e^{-\frac{1}{2}[A,B]} \dots \tag{7}$$

Then we insert complete set of momentum eigen states $|p\rangle$:

$$\begin{aligned}
\langle x_{i+1} | e^{-iH\delta t} | x_i \rangle &= \int \frac{dp_i}{2\pi} \langle x_{i+1} | e^{-i\delta t \frac{p_i^2}{2m}} | p_i \rangle \langle p_i | e^{-i\delta t V(x)} | x_i \rangle \\
&= \int \frac{dp_i}{2\pi} \exp \left[i p_i (x_{i+1} - x_i) - i \left(\frac{p_i^2}{2m} + V(x_i) \right) \delta t \right]
\end{aligned} \tag{8}$$

If we do this for each term in Equation 5, we will get:

$$\langle x' | e^{-iHt} | x \rangle = \int_{x(0)=x}^{x(t)=x'} \left(\prod_{i=1}^{N-1} dx_i \prod_{j=0}^{N-1} \frac{dp_j}{2\pi} \right) \exp \left[i \sum_{i=0}^{N-1} \left(p_i (x_{i+1} - x_i) - \left(\frac{p_i^2}{2m} + V(x_i) \right) \delta t \right) \right] \tag{9}$$

- **Step 3:** then we integrate out p_i . This is a gaussian integral, thus we can get:

$$\langle x' | e^{-iHt} | x \rangle = \int_{x(0)=x}^{x(t)=x'} \left(\prod_{i=1}^{N-1} dx_i f(\delta t, m) \right) \exp \left[i \sum_{i=0}^{N-1} \left(\frac{m}{2} \left(\frac{x_{i+1} - x_i}{\delta t} \right)^2 - V(x_i) \right) \delta t \right] \tag{10}$$

where $f(\delta t, m)$ is caused by the gaussian integral.

Remark

In some contexts eg. Lattice field theory, we mainly don't integrate out p_i and keep the path integral formula in terms of both x and p .

- **Step 4:** we take the limit of $N \rightarrow \infty$ and $\delta t \rightarrow 0$, then we can get the final path integral formula:

$$\langle x' | e^{-iHt} | x \rangle = \int_{x(0)=x}^{x(t)=x'} \mathcal{D}x \exp \left[i \int_0^t dt \left(\frac{m}{2} \dot{x}^2 - V(x) \right) \right] \tag{11}$$

Where we define the measure of the path integral as:

$$\mathcal{D}x = \lim_{N \rightarrow \infty} \left(\prod_{i=1}^{N-1} dx_i f(\delta t, m) \right) \tag{12}$$

Moreover, we notice that the exponent in the path integral is just the action of the system, thus we can write the path integral formula in a more compact way:

$$\langle x' | e^{-iHt} | x \rangle = \int_x^{x'} \mathcal{D}x \exp(iS[x]) \tag{13}$$

Remark

Notice that the path integral measure is in fact a mathematically ill-defined object. Yet to make it physical meaningful, we have to impose some physical conditions on it.

1.1.2 Correlation Functions as Functional Integral

Correlation functions are transition amplitudes with time ordered operator insertions, in Heisenberg picture, we can write the correlation functions as:

$$\langle x', t' | \mathcal{T} O(t_1) \dots O(t_n) | x, t \rangle \quad (14)$$

YL: [to be continued]

1.1.3 Generating Functional and Functional Derivative

Generating functional is a very useful tool to calculate correlation functions in quantum theory. We define a generating functional as the transition amplitude of a slightly modified theory:

$$H(p, x) \rightarrow H(p, x) - J(t)x(t) \quad (15)$$

If we repeat the same procedure as in the previous section, we can get the path integral formula for the generating functional:

$$Z[J] = \int_x^{x'} \mathcal{D}x \exp \left[i \int_0^t dt (L + J(t)x(t)) \right] \quad (16)$$

Then we can see that the correlation functions can be calculated by taking functional derivatives of the generating functional:

1.1.4 Euclidean path integral

1.2 Free Real Scalar Field Path Integral

1.3 Quantum Conservation and Ward identity

1.4 Dirac Field Path Integral

1.5 Non-Abelian Gauge Theory Path Integral

Bibliography

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- [2] P. Di Francesco, P. Mathieu, and D. Sénéchal, *Conformal Field Theory*. in Graduate Texts in Contemporary Physics. New York, NY: Springer New York, 1997. doi: [10.1007/978-1-4612-2256-9](https://doi.org/10.1007/978-1-4612-2256-9).